

Parallelization of Neural Network Training

Final presentation | High-Performance Big Data and Artificial Intelligence Systems (高效能巨量資料與人工智慧系)

Lukas Voss | 5 June 2023

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Agenda

Parallelization of Neural Network Training

1. **Motivation**
   - Growing impact of Artificial Intelligence (AI) makes efficient resource use crucial for advancements
   - GPU provides attractive parallelization capabilities for matrix-operations like in Neural Networks
2. **Implementation + Results**
   - CPU vs. GPU in like-for-like analysis for image classification with MNIST data set
   - Varying hyperparameters to investigate impact on performance and accuracy KPIs
3. **Summary and Outlook**
   - Key findings and highlights
   - Discussion and Q&A

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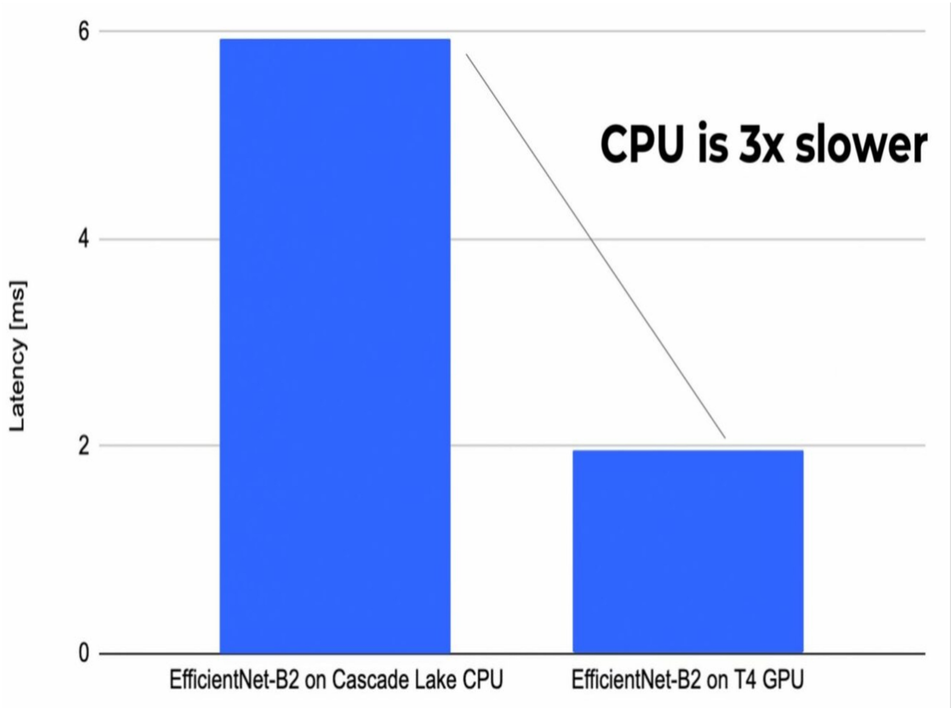
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Training of Neural Networks

CPUs are not well-suited to handle workload and lead to prohibitively long training times

The current performance gap [1]



Deep Learning's popularity >10x over the last decade [2]

- Research on deep learning models has accelerated due to increased data availability, better algorithms and hardware
- Especially in focus for AI-related tasks

- Text, object and speed recognition for NLP tasks
- Robotics, including autonomous vehicles and robotic control systems

- Widely available CPUs cannot efficiently process the amount of used data for training large models
- Time and resource constraints become important factors

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Deep Dive: Motivation

AI has the potential to lead to major efficiency improvements across industries

“

It's very clear that AI is going to impact every industry. I think that every nation needs to make sure that AI is a part of their national strategy. Every country will be impacted.



- Jensen Huang[1]

Healthcare



- Enhance diagnostics, e.g., medical imaging analysis
- Predict disease outcomes
- Enable personalized medicine, and facilitate drug discovery

Education



- Intelligent educational content generation
- Support personalized learning
- Adaptive tutoring systems, automated grading

1) Founder and CEO of NVIDIA; Quote taken from an interview in 2018

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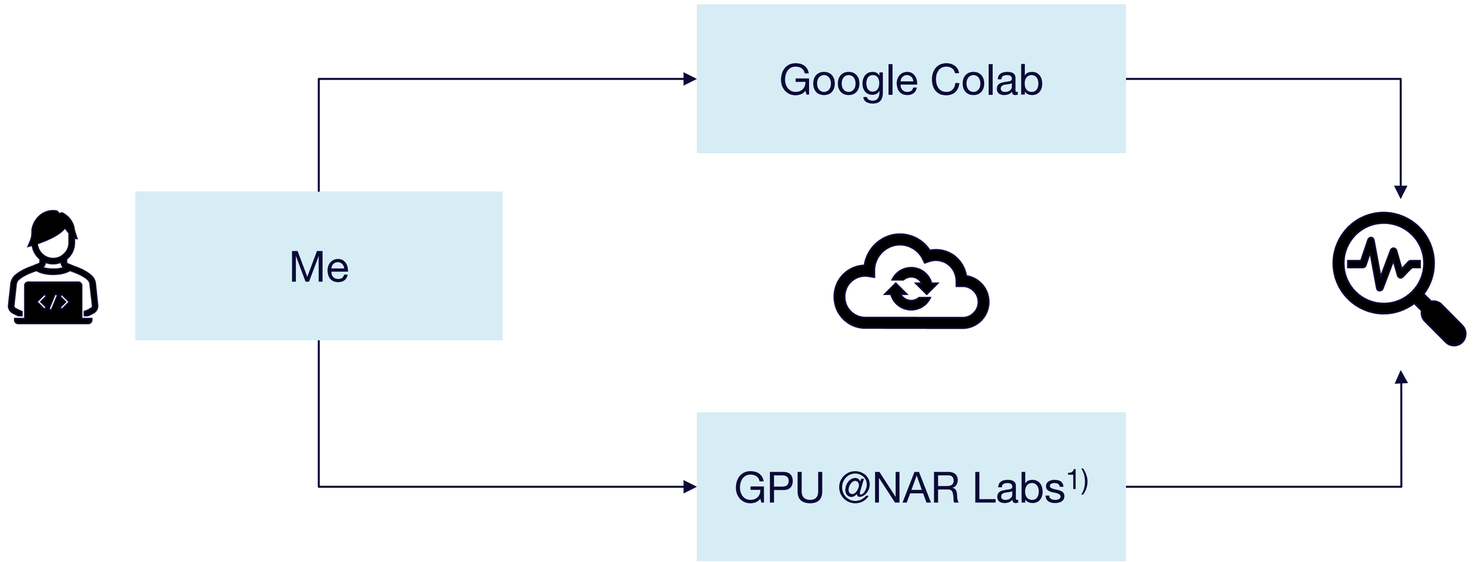
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System Architecture

Access to GPU was granted by Taiwan High-Performance Computing Cluster NAR Labs



System Design

- An account at NAR Labs enabled us to use their NVIDIA RTX 3070 GPU as a cloud-based service
- Convenient access on-demand for improved flexibility

1) National Center for High-Performance Computing

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Neural Network Training

A comprehensive approach to test performance potential lifted by GPU

Dataset



- Image recognition task with 28x28 pixel on a grayscale
- The MNIST dataset has handwritten digits with 60,000 training and 10,000 test images

Method



- Convolutional Neural Network with ReLu activation and pooling layer
- Softmax output layer

Evaluation and benchmarking again CPU



- Conduction like-for-like comparisons[1)] between Google Colab CPU and GPU
- GPU will be tested with and without explicit data loading parallelization

Note MLP: multi-layer perceptron **1)** Evaluation on the same task

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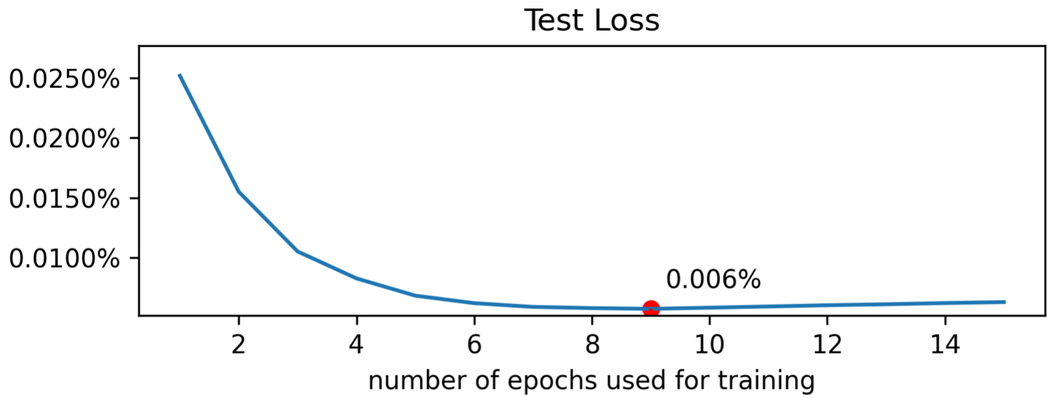
Google Colab CPU: Training epochs

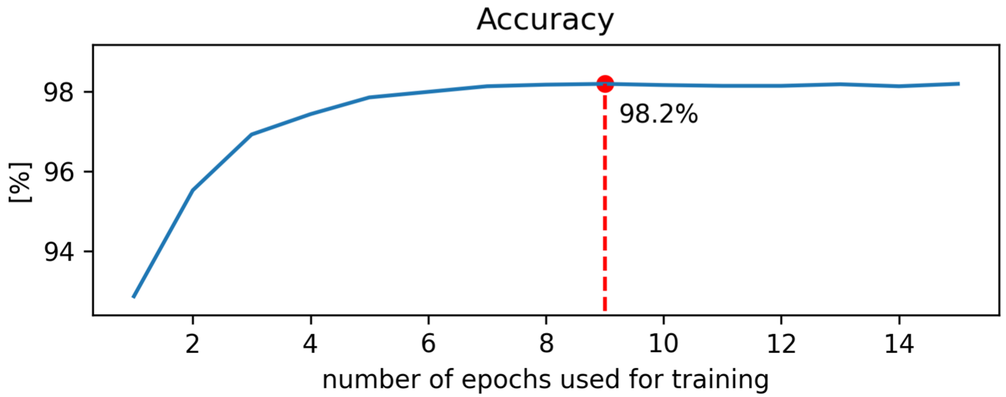
After a sweet spot of #epochs has been surpassed, more epochs do more harm than good

Parameters

Execution Time	48 sec
Optimizer	SGD[1]
Learning Rate	0.1

Performance





1) Stochastic Gradient Descent

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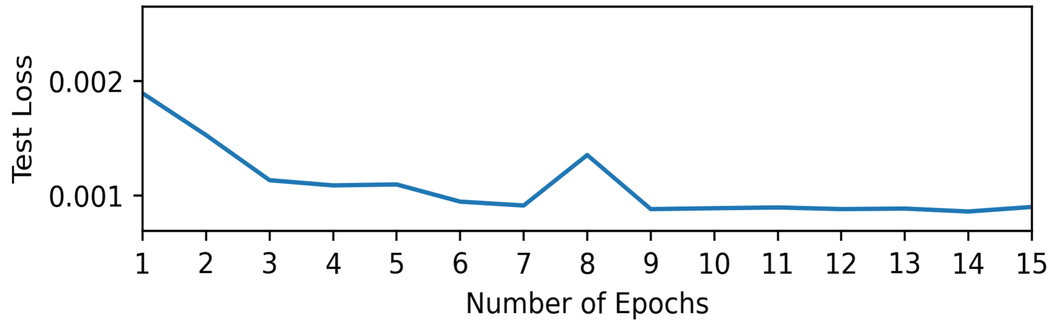
NVIDIA RTX 3070: Training epochs

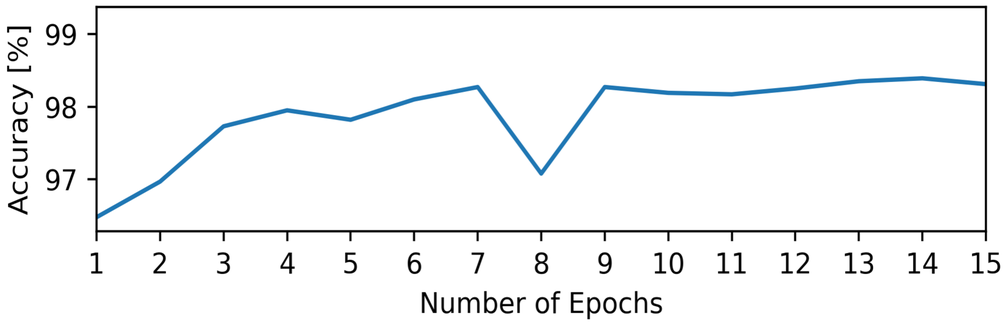
Data loading operations parallelized with four worker processes leads to performance gain

Parameters

Execution Time	16 sec
Optimizer	SGD[1]
Learning Rate	0.1

Performance





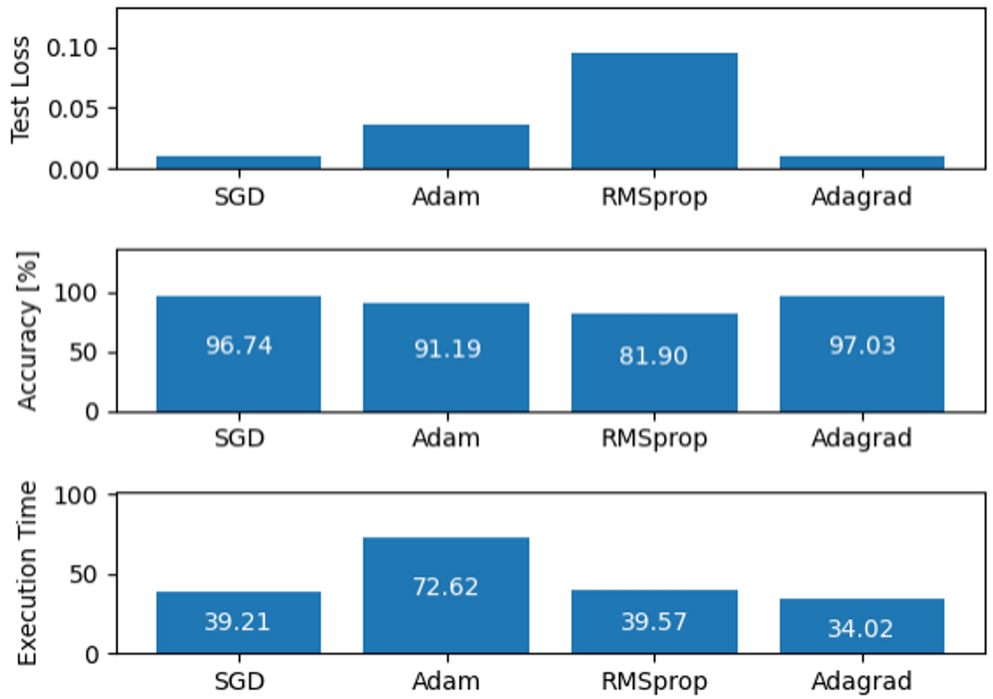
1) Stochastic Gradient Descent

 © Lukas Voss

Deep Dive: Variation of Optimizers

Leveraging the capabilities of the GPU, we achieve a significant speed up in training time

Google Colab CPU: Intel(R) Xeon(R) CPU @ 2.20GHz

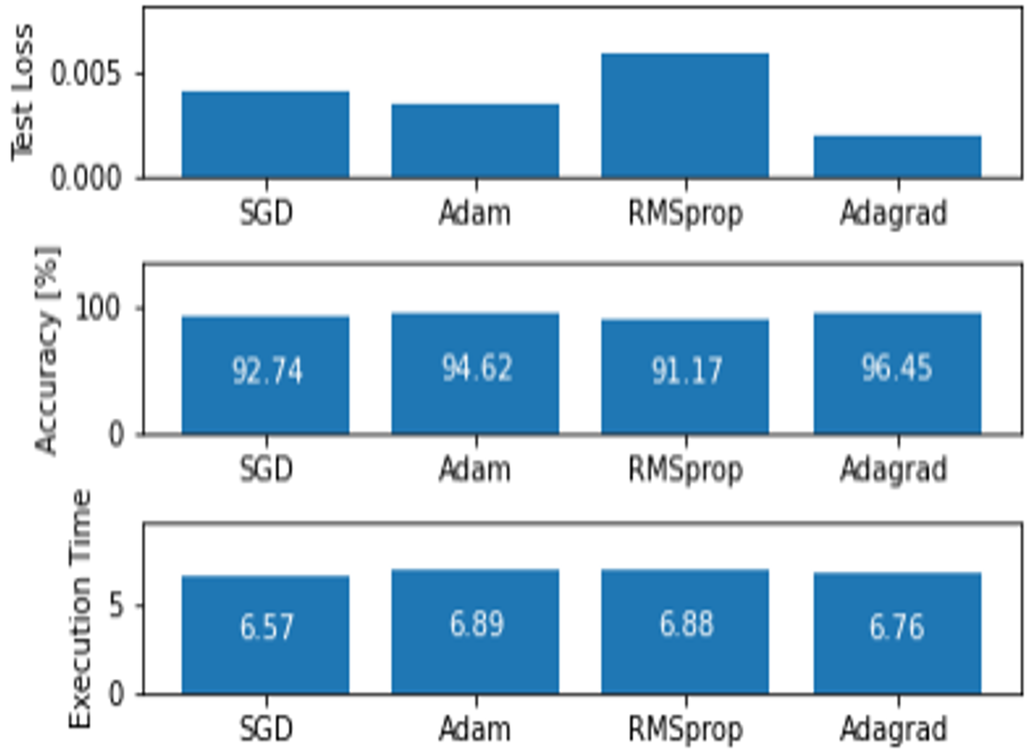


10x

similar

>6x

GPU: NVIDIA RTX 3070



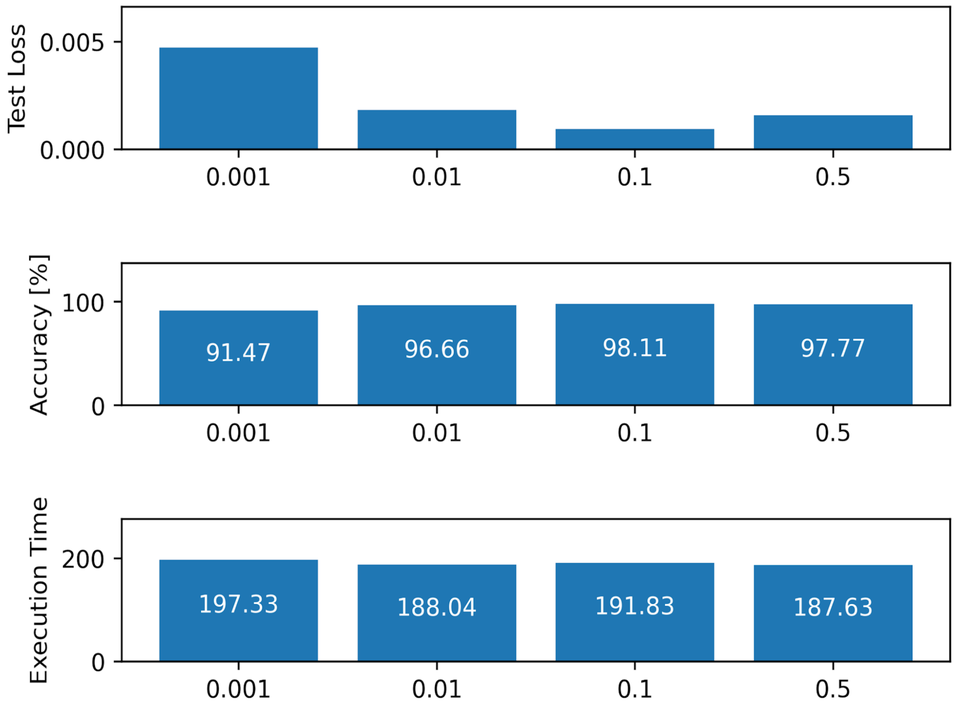
Improvement by GPU

 © Lukas Voss

Deep Dive: Learning Rates

Leveraging the capabilities of the GPU, we achieve a significant speed up in training time

Google Colab CPU: Intel(R) Xeon(R) CPU @ 2.20GHz

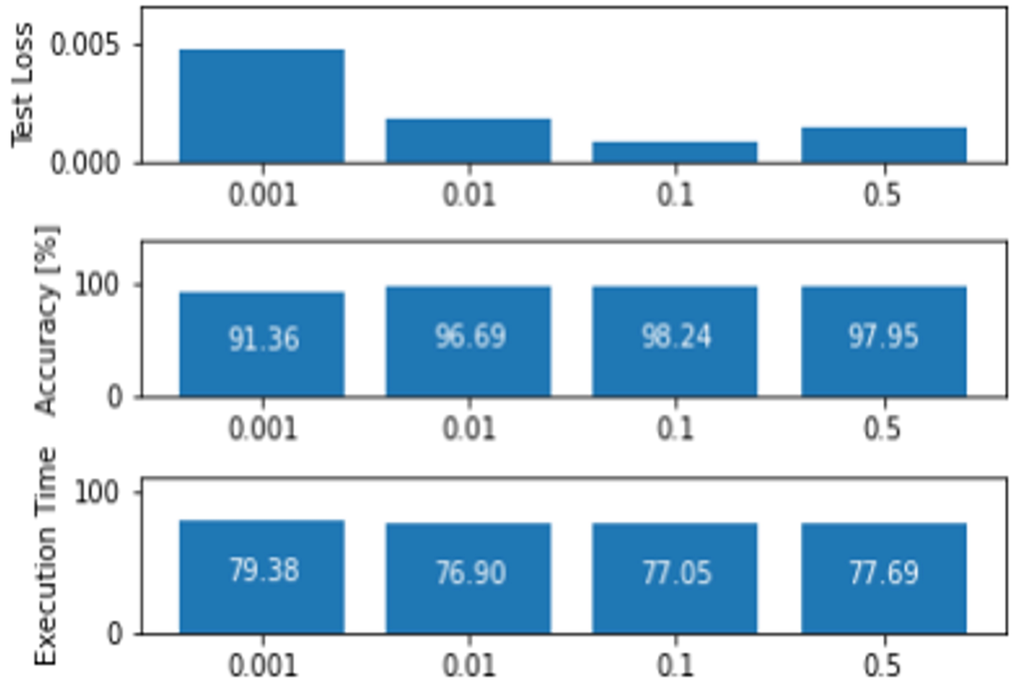


same

same

>2x

GPU: NVIDIA RTX 3070



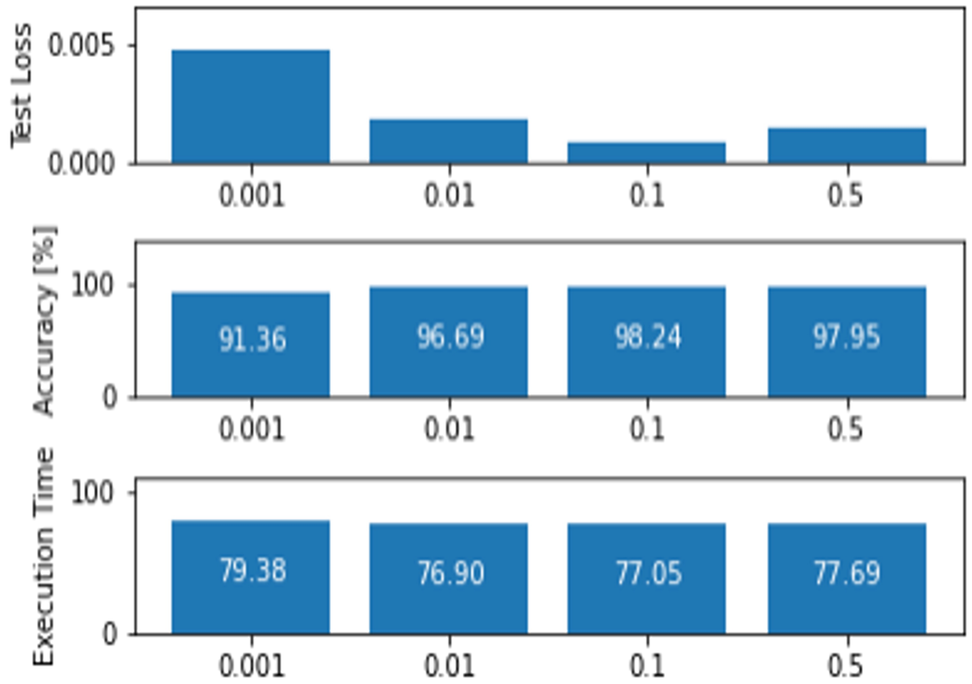
Improvement by GPU

 © Lukas Voss

Data Parallelization on GPU: Learning Rates

Data loading operations **parallelized with four worker processes** leads to performance gain

Without Data Parallelization: Learning Rates

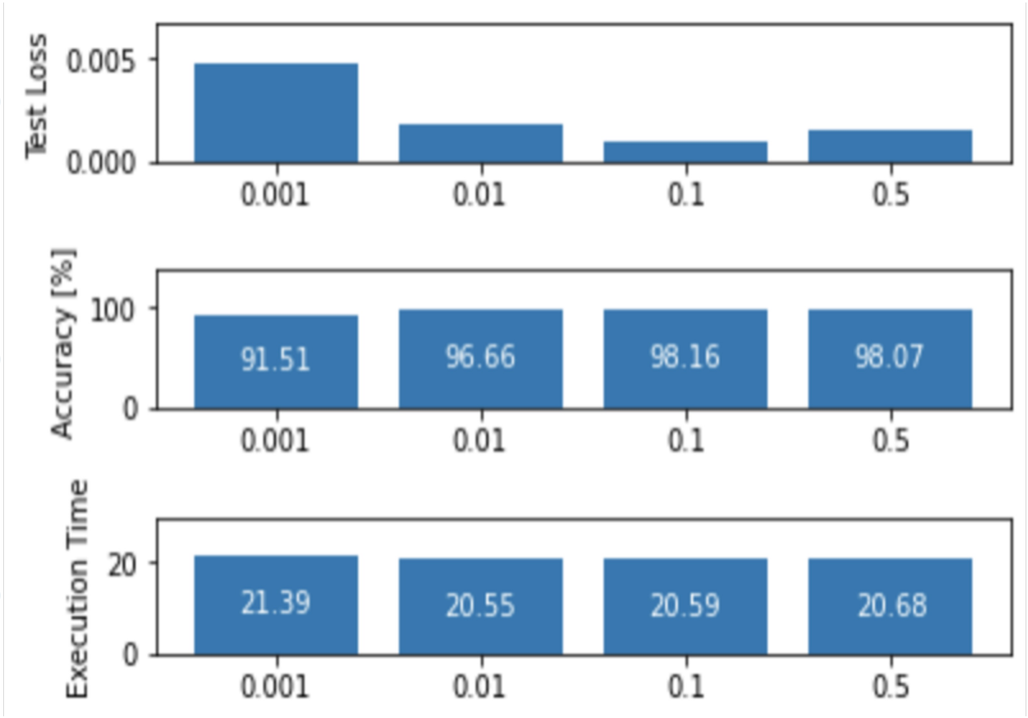


same

similar

ca. 4x

With Data Parallelization: Learning Rates



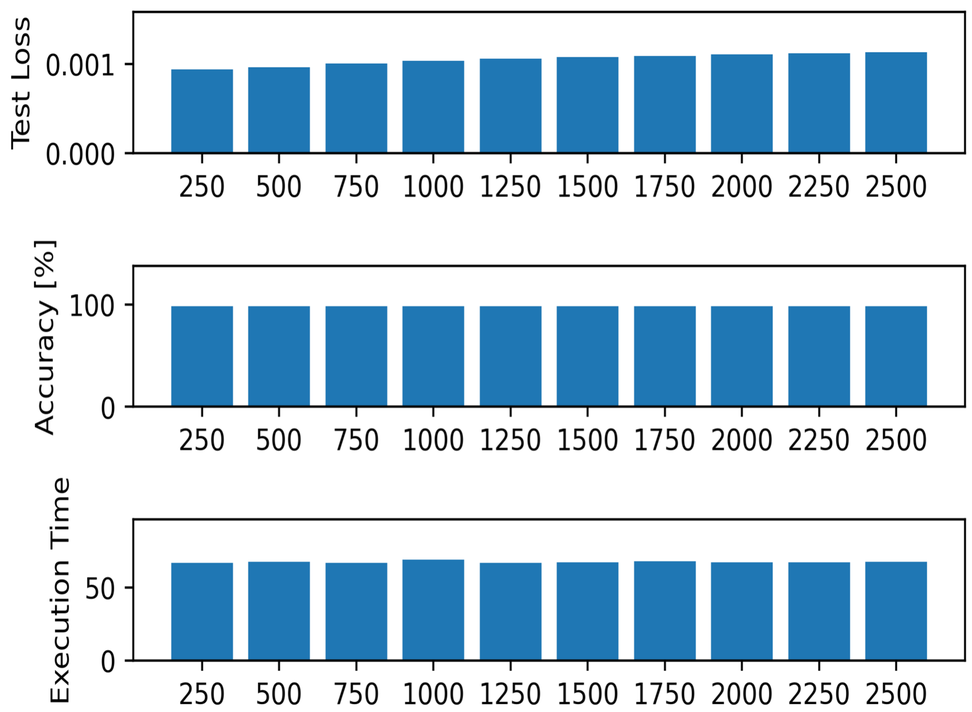
Improvement by GPU

 © Lukas Voss

Data Parallelization on GPU: Batch Size

Data loading operations **parallelized with four worker processes** leads to performance gain

Without Data Parallelization: Batch Size

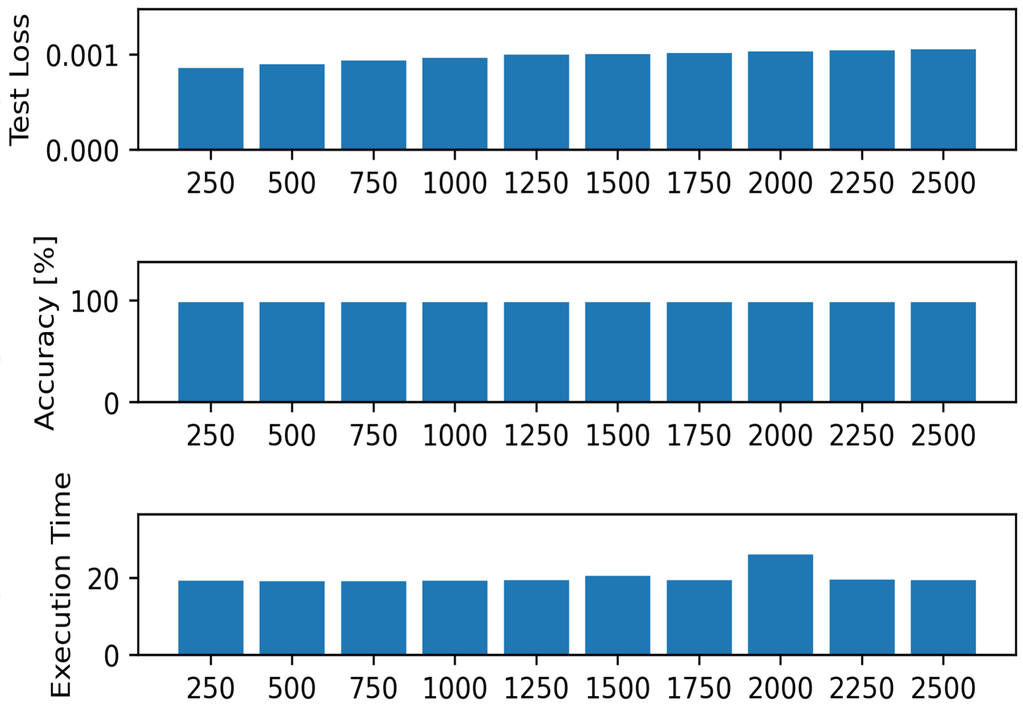


similar

similar

ca. 3x

With Data Parallelization: Batch Size



Improvement by GPU

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Executive Summary

Customizing the GPU‘s inbuilt capacity to parallelize further improves performance

Performance



- Significant **improvements** in **execution time up to 8x faster** compared to CPU
- Model has different sensitivity to different parameters

Distributed Architecture



- Efficiently load and preprocess data in mini-batches during the training and testing phases

Outlook



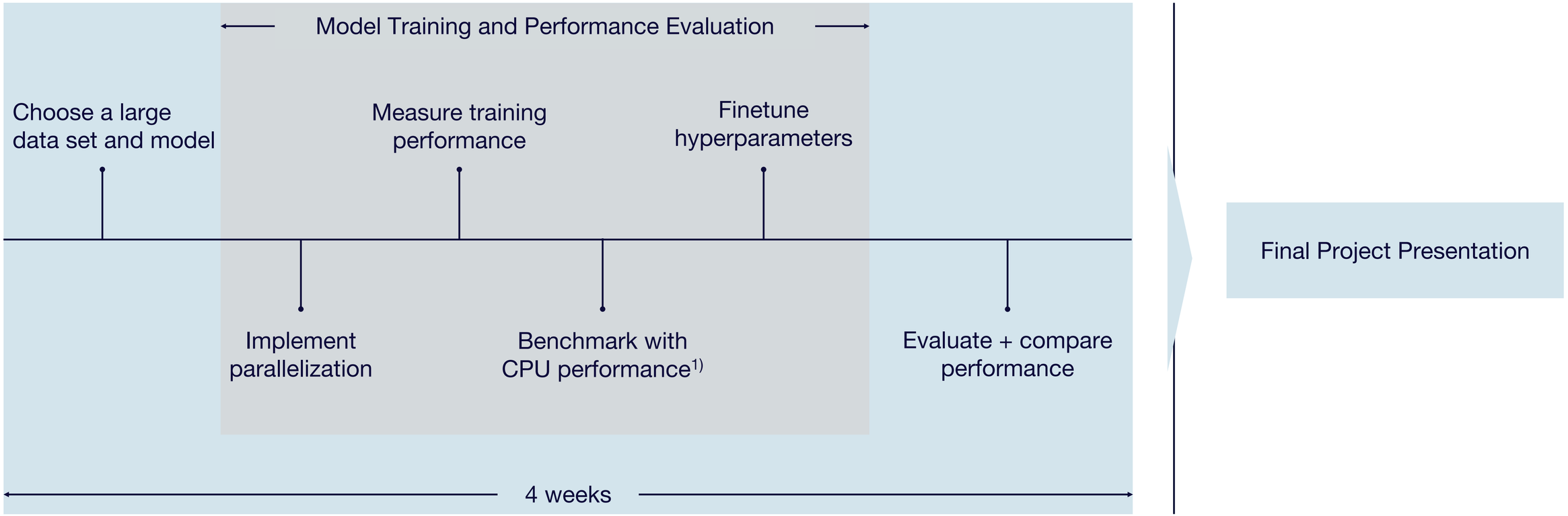
- Fine-tuning the implementation on the architecture
- Explore model parallelism
- Consult with industry experts and literature



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Appendix: Timeline of Implementation

Top-level view of the proposed research idea: Parallelization of Deep Learning Training



1) Google Colab CPU: Intel(R) Xeon(R) CPU @ 2.20GHz

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Appendix: Literature

Overview of used research articles and other sources

- [1]: Deci AI: Can You Close the Performance Gap Between GPU and CPU for Deep Learning Models?
- [2]: Md Jaber Al Nahian et al. „Camera Model Identification using Deep CNN and Transfer Learning Approach“ (2019)
- [3] M. Abadi et al. "Deep Learning on GPUs versus CPUs: A Performance Comparison" (2016)
- [4] Y. Liu et al. "Deep Learning on CPU and GPU: Performance and Energy Considerations" (2017)
- [5] D. Ciresan et al. "Accelerating Deep Convolutional Neural Networks using Specialized Hardware" (2018)
- [6] C. Zhang et al. "Comparison of Deep Learning Training on CPUs, GPUs, and FPGAs" (2019)
- [7] J. Li et al. "Accelerating Deep Learning via Speed-of-Light Neural Networks on GPUs and CPUs" (2020)

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